



LABORATORIO DE REDES DE COMPUTADORAS – INF238
LABORATORIO 04

TEMA: Enrutamiento Estático

DATOS:				
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Horario: 784				Fecha: 01/05/25
CALIFICACIÓN:				
Prueba de entrada: (5 ptos)	Trabajo Previo: (3 ptos)	Trabajo en Laboratorio: (10 ptos)	Evaluación Oral: (2 ptos)	Nota Final (20 ptos):
OBSERVACIONES:				

TRABAJO PREVIO

Configuraciones básicas en routers, rutas estáticas

Objetivos:

- ☐ Familiarizar al alumno con las configuraciones básicas en el Router para comprender cómo controlan el tráfico de red y su rol en la interconexión de dispositivos.
- ☐ Familiarizar al alumno con la configuración de rutas estáticas con el fin de optimizar y controlar el flujo del tráfico entre redes remotas.

Indicaciones:

- ☐ Resuelva claramente cada una de las preguntas y **evidencie cada paso con las imágenes respectivas, sino NO SE ASIGNARÁ EL PUNTAJE**
- ☐ Entregue la solución en archivo Packet Tracer.

Se recomienda que el alumno revise la parte teórica, ya que le servirá para desarrollar este trabajo y el laboratorio. **La prueba de entrada correspondiente será en base a la teoría y a este trabajo.**

Actividad # 1: Enrutamiento Estático.



Puntaje: 3.0 puntos

Implementar la siguiente red utilizando Packet Tracer para lograr conectividad entre las computadoras mediante enrutamiento estático en los routers. Definir las redes LAN y WAN que utilizará en base a lo brindado en la imagen 1, donde X corresponde al último dígito de su código. Usar Router modelo 2911 y Switch 2960. Para cada una de las redes LAN, deberá asignar las IPs correspondientes a las PCs.

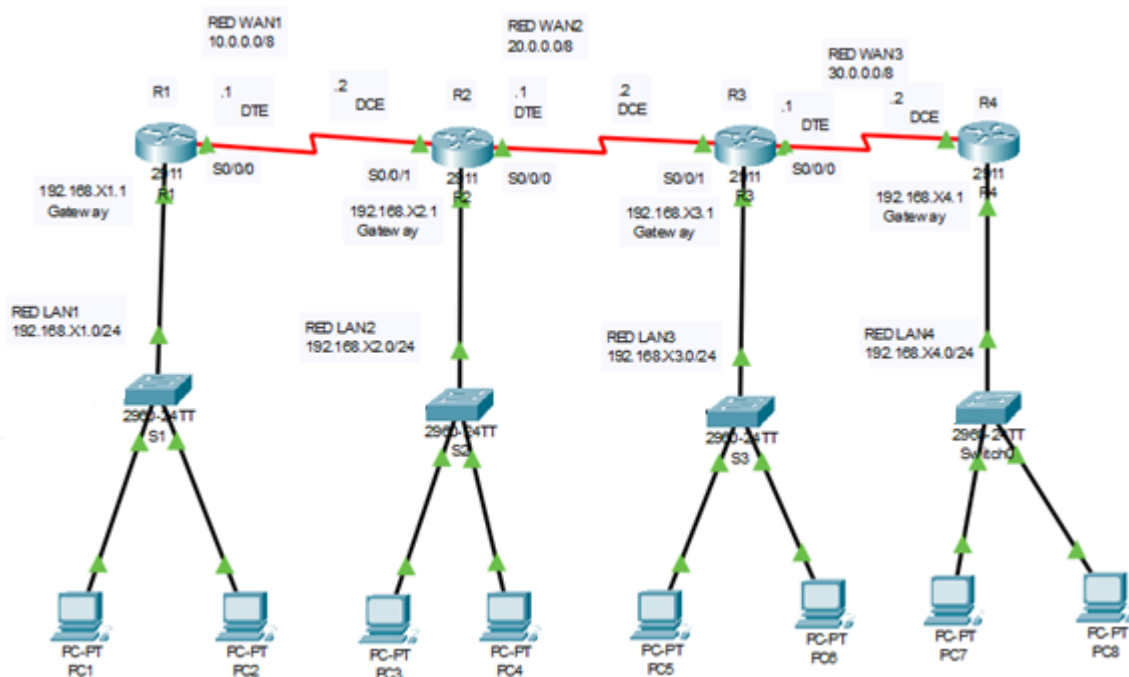
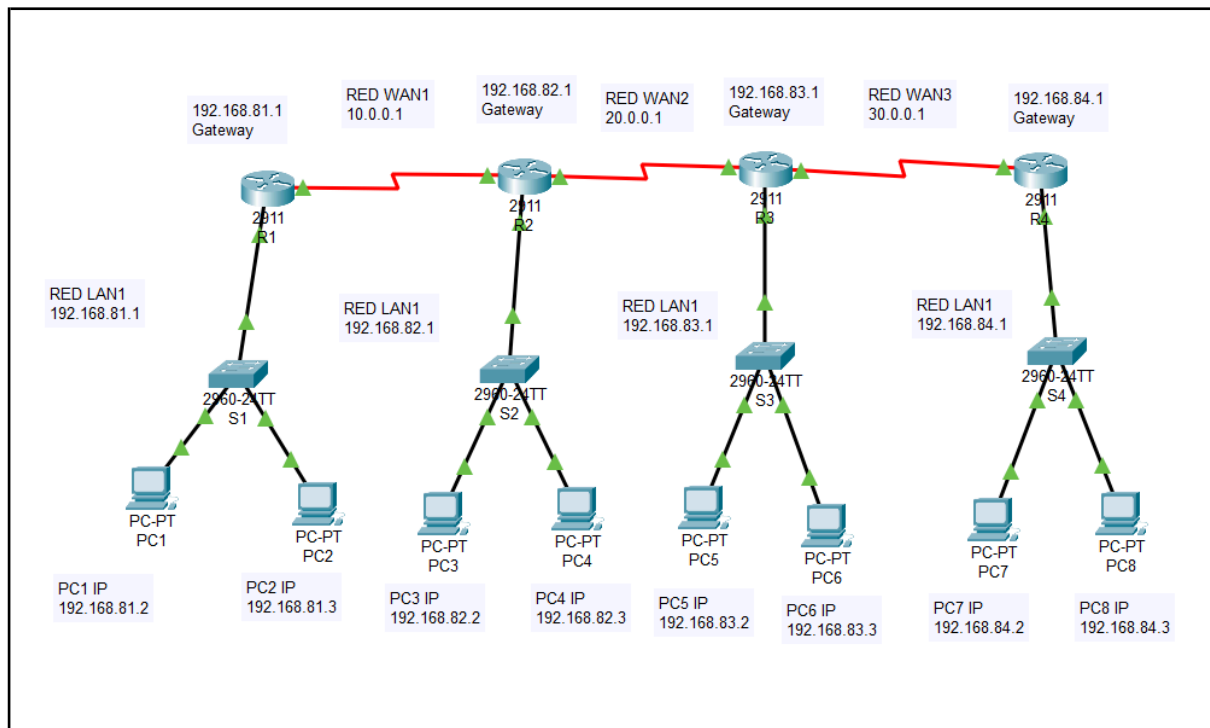


Imagen 1: Topología a implementar.

Se pide:

1. Presentar la topología de la red implementada en Packet Tracer. Indicar cada dispositivo con un nombre y su IP. (0.5 Pto.)



2. Configurar las interfaces de los routers y de las computadoras con las IP respectivas y las rutas estáticas. Presente todos los comandos usados en el Router 3. (0.5 Pto.)

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface s0/0/1
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
Router(config-if)#exit
Router(config)#
```

```

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/0/0
Router(config-if)# ip address 30.0.0.1 255.0.0.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
exit
Router(config)#ip route 192.168.81.0 255.255.255.0 20.0.0.1
Router(config)#ip route 192.168.82.0 255.255.255.0 20.0.0.1
Router(config)#ip route 192.168.84.0 255.255.255.0 30.0.0.2
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```

3. Presentar captura de imágenes de la salida del comando **show ip interface brief** de los cuatro routers. (0.5 Pto.)

R1:

```

Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 192.168.81.1    YES manual up              up
GigabitEthernet0/1 unassigned      YES unset  administratively down down
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Serial0/0/0        10.0.0.1        YES manual up              up
Serial0/0/1        unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down

```

R2:

```

Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 192.168.82.1    YES manual up              up
GigabitEthernet0/1 unassigned      YES unset  administratively down down
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Serial0/0/0        20.0.0.1        YES manual up              up
Serial0/0/1        10.0.0.2        YES manual up              up
Vlan1              unassigned      YES unset  administratively down down

```

R3:

```

Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 192.168.83.1    YES manual up              up
GigabitEthernet0/1 unassigned      YES unset  administratively down down
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Serial0/0/0        30.0.0.1        YES manual up              up
Serial0/0/1        20.0.0.2        YES manual up              up
Vlan1              unassigned      YES unset  administratively down down

```

R4:

```
Router# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.84.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/0/0	unassigned	YES	unset	administratively down	down
Serial0/0/1	30.0.0.2	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

4. Presentar captura de imágenes de la tabla de rutas de los cuatro routers con el comando **show ip route**. (0.5 Pto.)

R1:

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
192.168.81.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.81.0/24 is directly connected, GigabitEthernet0/0
L       192.168.81.1/32 is directly connected, GigabitEthernet0/0
S       192.168.82.0/24 [1/0] via 10.0.0.2
S       192.168.83.0/24 [1/0] via 10.0.0.2
S       192.168.84.0/24 [1/0] via 10.0.0.2
```

R2:

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Serial0/0/1
L       10.0.0.2/32 is directly connected, Serial0/0/1
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Serial0/0/0
L       20.0.0.1/32 is directly connected, Serial0/0/0
S       192.168.81.0/24 [1/0] via 10.0.0.1
192.168.82.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.82.0/24 is directly connected, GigabitEthernet0/0
L       192.168.82.1/32 is directly connected, GigabitEthernet0/0
S       192.168.83.0/24 [1/0] via 20.0.0.2
S       192.168.84.0/24 [1/0] via 20.0.0.2
```

R3:

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Serial0/0/1
L       20.0.0.2/32 is directly connected, Serial0/0/1
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Serial0/0/0
L       30.0.0.1/32 is directly connected, Serial0/0/0
S       192.168.81.0/24 [1/0] via 20.0.0.1
S       192.168.82.0/24 [1/0] via 20.0.0.1
    192.168.83.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.83.0/24 is directly connected, GigabitEthernet0/0
L       192.168.83.1/32 is directly connected, GigabitEthernet0/0
S       192.168.84.0/24 [1/0] via 30.0.0.2

```

R4:

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Serial0/0/1
L       30.0.0.2/32 is directly connected, Serial0/0/1
S       192.168.81.0/24 [1/0] via 30.0.0.1
S       192.168.82.0/24 [1/0] via 30.0.0.1
S       192.168.83.0/24 [1/0] via 30.0.0.1
    192.168.84.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.84.0/24 is directly connected, GigabitEthernet0/0
L       192.168.84.1/32 is directly connected, GigabitEthernet0/0

```

5. Presentar captura de imágenes de (1.0 Pto.):

Prueba de ping y traceroute entre PC3 y PC8.

Prueba de ping y traceroute entre PC2 y PC5.

Prueba de ping y traceroute entre PC6 y el Gateway de PC1.

Prueba de ping y traceroute entre PC4 y el Gateway de PC7.

Prueba de ping y traceroute entre PC3 y PC8:

ping:

```
C:\>ping 192.168.84.3

Pinging 192.168.84.3 with 32 bytes of data:

Reply from 192.168.84.3: bytes=32 time=22ms TTL=125
Reply from 192.168.84.3: bytes=32 time=19ms TTL=125
Reply from 192.168.84.3: bytes=32 time=23ms TTL=125
Reply from 192.168.84.3: bytes=32 time=20ms TTL=125

Ping statistics for 192.168.84.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 23ms, Average = 21ms
```

traceroute:

```
C:\>tracert 192.168.84.3

Tracing route to 192.168.84.3 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.82.1
  1  5 ms    2 ms    0 ms    20.0.0.2
  2  1 ms    7 ms    1 ms    30.0.0.2
  3  4 ms    9 ms    5 ms    192.168.84.3

Trace complete.
```

Prueba de ping y traceroute entre PC2 y PC5:

ping:

```
C:\>ping 192.168.83.2

Pinging 192.168.83.2 with 32 bytes of data:

Reply from 192.168.83.2: bytes=32 time=20ms TTL=125
Reply from 192.168.83.2: bytes=32 time=20ms TTL=125
Reply from 192.168.83.2: bytes=32 time=2ms TTL=125
Reply from 192.168.83.2: bytes=32 time=23ms TTL=125

Ping statistics for 192.168.83.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 23ms, Average = 16ms
```

traceroute:

```
C:\>tracert 192.168.83.2

Tracing route to 192.168.83.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.81.1
  2  7 ms      0 ms      8 ms      10.0.0.2
  3  8 ms      7 ms      8 ms      20.0.0.2
  4  1 ms      4 ms      5 ms      192.168.83.2

Trace complete.
```

Prueba de ping y traceroute entre PC6 y el Gateway de PC1:

ping:

```
C:\>ping 192.168.81.1

Pinging 192.168.81.1 with 32 bytes of data:

Reply from 192.168.81.1: bytes=32 time=2ms TTL=253
Reply from 192.168.81.1: bytes=32 time=20ms TTL=253
Reply from 192.168.81.1: bytes=32 time=14ms TTL=253
Reply from 192.168.81.1: bytes=32 time=19ms TTL=253

Ping statistics for 192.168.81.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 20ms, Average = 13ms
```

traceroute:

```
C:\>tracert 192.168.81.1

Tracing route to 192.168.81.1 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.83.1
  2  8 ms      0 ms      0 ms      20.0.0.1
  3  1 ms      8 ms      4 ms      192.168.81.1

Trace complete.
```

Prueba de ping y traceroute entre PC4 y el Gateway de PC7:

ping:

```
C:\>ping 192.168.84.1

Pinging 192.168.84.1 with 32 bytes of data:

Reply from 192.168.84.1: bytes=32 time=22ms TTL=253
Reply from 192.168.84.1: bytes=32 time=23ms TTL=253
Reply from 192.168.84.1: bytes=32 time=2ms TTL=253
Reply from 192.168.84.1: bytes=32 time=19ms TTL=253

Ping statistics for 192.168.84.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 23ms, Average = 16ms
```

tracert:

```
C:\>tracert 192.168.84.1

Tracing route to 192.168.84.1 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.82.1
  2  0 ms      6 ms      4 ms      20.0.0.2
  3  5 ms      1 ms      0 ms      192.168.84.1

Trace complete.
```

6. Adjuntar su archivo de Packet Tracer con las configuraciones completas.

Nota:

Los Routers en el PT por defecto no cuentan con las interfaces seriales S0/0/0 y S0/0/1, es necesario instalar las tarjetas HWIC, para ello debe apagar el Router y agregar el módulo, luego encender el Router.

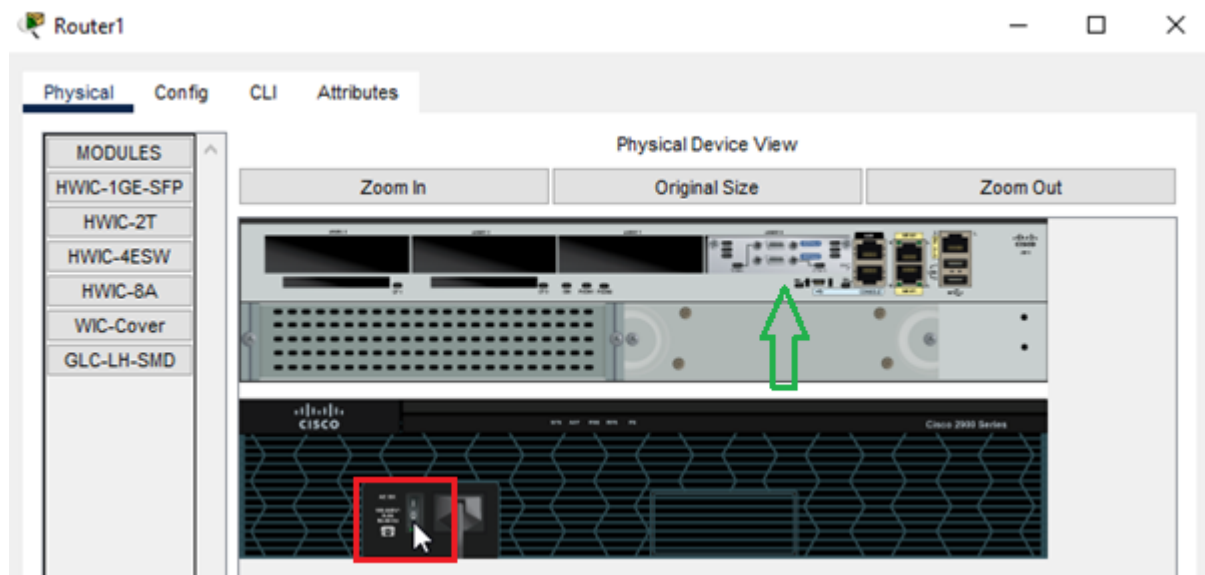


Imagen 2: Apagando/Encendiendo el Router y agregando el módulo HWIC-2T.